

RUO'ERGA, China

WMO: 560790

Lat: 33.5748N

Lon: 102.9660E

Elev: 11289

StdP: 9.61

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.5	(c) -1.3	(d) -17.1	(e) 3.3	(f) 12.7	(g) -13.8	(h) 4.0	(i) 13.4	(j) 21.6	(k) 29.8	(l) 18.7	(m) 31.2	(n) 2.0	(o) 40	(p) 0.701

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	69.5	53.9	67.1	52.8	64.6	51.6	56.2	66.3	54.9	64.2	53.5	61.8	5.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.5	90.3	61.1	51.3	86.3	59.5	50.2	82.8	58.2	29.5	66.7	28.4	64.4	27.4	62.0	61.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	15.5	13.3	-12.7	74.5	3.8	1.9	-15.5	75.9	-17.7	77.0	-19.8	78.1	-22.5	79.5
			-13.4	58.8	3.7	1.3	-16.1	59.8	-18.2	60.5	-20.3	61.2	-22.9	62.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.2	17.8	22.7	30.7	38.4	44.1	49.6	53.9	53.3	47.9	38.6	28.5	20.0		
	DBStd	13.60	6.25	6.80	6.06	5.63	4.72	4.48	4.29	4.63	5.45	5.21	6.00	5.72		
	HDD50	5028	998	763	598	349	190	59	14	20	106	354	646	931		
	HDD65	10146	1463	1183	1063	798	648	461	345	362	514	817	1096	1396		
	CDD50	358	0	0	0	1	8	48	134	123	42	2	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	3	0	0	0	0	0	0	1	2	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	5.3	5.1	5.8	6.2	6.2	6.0	5.5	5.0	4.8	5.1	4.9	4.5	4.5		
(15) Precipitation	PrecAvg	26.0	0.2	0.3	0.7	1.5	3.1	3.8	5.0	4.4	4.6	2.2	0.3	0.1		
	PrecMax	34.0	0.4	0.8	1.1	3.0	4.6	7.6	8.9	7.8	8.4	5.7	0.8	0.3		
	PrecMin	19.4	0.0	0.0	0.3	0.6	1.9	2.4	2.4	2.1	2.0	0.4	0.0	0.0		
	PrecStd	3.9	0.1	0.2	0.2	0.5	0.8	1.3	1.5	1.5	1.9	1.3	0.2	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.8	52.1	56.6	63.4	66.3	68.8	72.6	72.8	70.2	62.7	52.3	47.2		
		MCWB	29.2	32.1	35.7	41.6	47.0	53.0	55.6	55.8	53.4	47.7	34.5	31.0		
	2%	DB	41.4	46.4	51.3	58.1	61.8	64.8	69.4	69.4	65.9	56.5	48.1	41.7		
		MCWB	26.7	29.3	33.6	39.6	45.1	51.1	54.3	54.5	51.7	42.5	32.9	27.7		
	5%	DB	37.3	42.3	47.2	54.4	58.6	62.0	66.9	66.6	62.0	52.6	44.8	37.9		
		MCWB	24.6	27.9	32.0	38.0	43.7	49.6	53.3	53.2	50.0	40.5	31.5	25.5		
	10%	DB	33.3	38.0	43.1	50.8	55.3	59.4	64.2	63.9	58.1	49.1	41.6	34.4		
		MCWB	22.7	26.0	30.5	37.0	42.7	48.4	52.5	52.0	47.9	38.8	30.0	23.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.0	32.9	37.6	44.1	50.7	56.4	58.1	57.7	55.9	49.7	36.8	31.6		
		MCDB	45.1	50.1	52.7	57.8	61.4	65.5	68.7	69.2	66.2	61.3	48.6	46.1		
	2%	WB	27.2	30.2	35.0	41.6	47.9	53.9	56.3	56.1	53.3	45.4	34.5	28.4		
		MCDB	40.8	43.7	47.7	54.0	57.6	61.3	66.2	66.7	63.1	53.0	45.2	40.5		
	5%	WB	25.1	28.4	33.1	39.7	45.8	52.0	55.0	54.7	51.5	42.6	32.4	26.1		
		MCDB	36.6	40.8	44.6	51.0	55.1	59.0	64.1	64.4	60.0	49.6	42.1	36.7		
	10%	WB	23.0	26.5	31.3	37.9	43.9	50.1	53.6	53.3	49.5	40.3	30.7	24.2		
		MCDB	32.7	36.9	40.9	48.1	52.7	56.7	61.5	61.7	56.2	46.7	39.5	33.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	29.3	27.3	21.8	21.0	19.9	17.4	18.9	19.5	18.5	18.9	24.0	28.1		
		MCDBR	36.0	35.1	28.9	28.7	26.5	22.5	24.3	24.2	24.4	25.2	28.7	33.3		
		MCWBR	23.2	20.4	13.8	12.2	11.9	10.5	11.1	11.0	11.6	12.5	15.1	20.8		
		MCDBR	33.3	31.6	24.9	24.0	21.6	17.0	19.6	19.9	20.3	20.1	25.2	30.8		
	5% WB	MCWBR	21.7	18.7	12.3	11.3	11.0	8.7	9.6	9.4	9.9	10.9	13.9	19.5		
(40) Clear-Sky Solar Irradiance	taub		0.242	0.274	0.333	0.353	0.345	0.337	0.337	0.338	0.312	0.276	0.243	0.223		
	taud		2.471	2.383	2.200	2.196	2.311	2.412	2.434	2.419	2.488	2.546	2.550	2.558		
	Ebn at Noon		315	312	298	296	298	298	298	297	303	308	313	319		
	Edh at Noon		27	33	43	46	41	37	36	36	32	28	25	23		
(44) All-Sky Solar Radiation	RadAvg		1088	1295	1547	1718	1721	1612	1756	1645	1334	1210	1131	1033		
	RadStd		53	76	72	86	77	147	180	180	112	120	41	45		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.30	+2.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.63	N/A	N/A	N/A	N/A	N/A	-77	-171	+101

Nomenclature: See separate page